σ-Algebra.

Def. Let X be a set.
A collection $\mathcal{M} \subseteq \mathcal{P}(X)$ is called a σ-algebra in X if:
1) $X \in \mathcal{M}$.
2). If $A \in \mathcal{M}$, then $A^c \in \mathcal{M}$
3). If $A = \bigcup_{n=1}^{\infty} A_n$ and if $A_n \in \mathcal{M}$ for $n \in \mathbb{N}$, then $A \in \mathcal{M}$.

For the such a σ-algebra $\mathcal{M}$, $(X, \mathcal{M})$ is called a mesurable space.
And, an element of $\mathcal{M}$ is called a measurable set.

Def. Let $(X, \mathcal{M})$ be a measurable space and
$(Y, \mathcal{L})$ be a topological space.
A function $f: X \to Y$ is called a measurable if:
for any open set $U \subseteq Y$, $f^{-1}[U] \subseteq X$ is a measurable set.
That is, $U \in \mathcal{L} \Rightarrow f^{-1}[U] \in \mathcal{M}$.

Thm. Let Y and Z be topological spaces, $g: Y \to Z$ be a Continuous.
If X is a measurable space and $f: X \to Y$ is a measurable,
then $g \circ f$ is a measurable.
Proof. Given an open set $U \subseteq Z$, $g^{-1}[U] \subseteq Y$ is open, so
$(g \circ f)^{-1}[U] = f^{-1}[g^{-1}[U]]$ is measurable. ☐

Borel set.

Thm. Let X be a set.
For any collection $\mathcal{F} \subseteq \mathcal{P}(X)$ of X,
there exists a smallest σ-algebra $\mathcal{M}^*$ containing $\mathcal{F}$ in X.

Proof. Let $\mathcal{C}$ be the family of all σ-algebras which contain $\mathcal{F}$.
That is, $\mathcal{C} = \{\mathcal{M} \subseteq \mathcal{P}(X) \mid \mathcal{M} \text{ is } \sigma\text{-algebra in } X, \text{ and } \mathcal{F} \subseteq \mathcal{M}\}$.

Since $\mathcal{P}(X)$ is a σ-algebra in X itself and contains $\mathcal{F}$,
therefore $\mathcal{C}$ is non-empty set. Define

$$\mathcal{M}^* = \bigcap_{\mathcal{M} \in \mathcal{C}} \mathcal{M}.$$

Clearly $\mathcal{F} \subseteq \mathcal{M}^*$, and $\mathcal{M}^* \subseteq \mathcal{M}$ for any $\mathcal{M} \in \mathcal{C}$.
Now, enough to show that $\mathcal{M}^*$ is a σ-algebra in X.
1) Since $X \in \mathcal{M}$ for all $\mathcal{M} \in \mathcal{C}$, $X \in \mathcal{M}^*$.
2). If $A \in \mathcal{M}^*$, then $A^c \in \mathcal{M}$ for all $\mathcal{M} \in \mathcal{C}$, thus $A^c \in \mathcal{M}^*$.
3). Assume $A_n \in \mathcal{M}^*$ $(n \in \mathbb{N})$. Then, for any $\mathcal{M} \in \mathcal{C}$, $A_n \in \mathcal{M}$
for all $n \in \mathbb{N}$, and so that $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$. Thus, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}^*$. ∎

Def. Let $(X, \mathcal{U})$ be a topological space.
The σ-algebra $\mathcal{B}$ in X generated by $\mathcal{U}$ is called the
Borel σ-algebra, and its element is called a Borel set.
The map $f: (X, \mathcal{U}) \to (Y, \mathcal{U}')$ is said to be Borel-Measurable if:
f is measurable with respects to $(X, \mathcal{M}_{\mathcal{U}})$,
where $\mathcal{M}_{\mathcal{U}}$ is Borel σ-algebra generated by $\mathcal{U}$.

Note: In Topology, F_σ is a set of all countable union of closed set,
G_δ is a set of all countable intersection of opens.

Lemma. Let $(X, \mathcal{M})$ be a measurable space,
$(Y, \mathcal{O})$ be a topological space.
And $f: X \to Y$ be a map. Then, a set

$$\Omega = \{E \subseteq Y \mid f^{-1}[E] \in \mathcal{M}\} \text{ is a } \sigma\text{-algebra in } Y.$$

Prof. Since $f^{-1}[Y] = X \in \mathcal{M}$, $Y \in \mathcal{O}$. And, given $A \in \mathcal{O}$, $f^{-1}[A] \in \mathcal{M}$, so $X \setminus f^{-1}[A] = f^{-1}[Y \setminus A] \in \mathcal{M}$, thus $Y \setminus A \in \mathcal{O}$. And, suppose $A_n \in \mathcal{O}$. Then, $f^{-1}[A_n] \in \mathcal{M}$ and so that $\bigcup_{n\in\mathbb{N}} f^{-1}[A_n] = f^{-1}\left[\bigcup_{n\in\mathbb{N}} A_n\right] \in \mathcal{M}$, it means $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{O}$. □

Thm. Let $(X, \mathcal{M})$ and $(Y, \mathcal{O})$ be measurable and topological space, respectively. Let $\mathcal{B}$ be the Borel σ-algebra in Y. Then,

If $f: X \to Y$ is measurable, then $f^{-1}[B] \in \mathcal{M}$ for any $B \in \mathcal{B}$.

Proof. Let $\Omega := \{E \subseteq Y \mid f^{-1}[E] \in \mathcal{M}\}$. By definition,

f is measurable $\Leftrightarrow \forall U \in \mathcal{O}$, $f^{-1}[U] \in \mathcal{M} \Leftrightarrow \mathcal{O} \subseteq \Omega$.

And, since the set Ω is a σ-algebra in Y by the lemma above, and the Borel σ-algebra $\mathcal{B}$ is the smallest σ-algebra containing $\mathcal{O}$, therefore $\mathcal{O} \subseteq \mathcal{B} \subseteq \Omega$. So proved. □

Thm. Let $f: X \to Y$ be a measurable, and $g: Y \to Z$ be a Borel map. Then, $g \circ f: X \to Y \to Z$ is a measurable.

Prof. Let $U \subset Z$ be an open set. By definition, $g^{-1}[U]$ is Borel set, and by the theorem above,

$$f^{-1}[g^{-1}[U]] = (g \circ f)^{-1}[U] \text{ is measurable in } X. \quad \square$$

Note: Since the Borel σ-algebra contains the topology, every continuous map is Borel map. But the converse is not true:

$f: \mathbb{R} \to \mathbb{R}; x \mapsto \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases}$, and Dirichlet $\chi_{\mathbb{Q}}(x)$.

Thm. Let f and g be real measurable functions on $(X, \mathcal{M})$. If $\varphi : [-\infty, \infty]^2 \to Y$ is a continuous map, then

$$\widehat{\varphi} : X \to Y : x \mapsto \varphi(f(x), g(x))$$

is measurable.

Proof. Let $h(x) = (f(x), g(x))$ for each $x \in X$, so that $\widehat{\varphi} = \varphi \circ h$. Let $I_1 \times I_2 \subseteq [-\infty, \infty]^2$ be given where $I_1, I_2 \subseteq [-\infty, \infty]$ are opens. Then,

$$h^{-1}[I_1 \times I_2] = f^{-1}[I_1] \cap g^{-1}[I_2],$$ measurable set, being intersection of m.s.

Now, given open set $V \subseteq [-\infty, \infty]^2$, which is second-countable, V is a countable union of $B_i = I_{i,1} \times I_{i,2}$. Therefore,

$$h^{-1}[V] = h^{-1}\left[\bigcup_{i=1}^{\infty} B_i\right] = \bigcup_{i=1}^{\infty} h^{-1}[B_i]$$ is a measurable set.

Consequently, $\widehat{\varphi}$ is measurable, being composition of cont'i. and m.s. ☑

Corollary. If $f, g : X \to [-\infty, \infty]$ are measurable, then

$$f+g, \quad fg, \quad f/g \text{ (if } g(x) \neq 0 \text{ for all } x \in X)$$

are measurable.